

Fruit and vegetable intake and overall cancer risk in the European Prospective Investigation into Cancer and Nutrition (EPIC).

BACKGROUND: It is widely believed that cancer can be prevented by high intake of fruits and vegetables. However, inconsistent results from many studies have not been able to conclusively establish an inverse association between fruit and vegetable intake and overall cancer risk.

METHODS: We conducted a prospective analysis of the European Prospective Investigation into Cancer and Nutrition (EPIC) cohort to assess relationships between intake of total fruits, total vegetables, and total fruits and vegetables combined and cancer risk during 1992-2000. Detailed information on the dietary habit and lifestyle variables of the cohort was obtained. Cancer incidence and mortality data were ascertained, and hazard ratios (HRs) and 95% confidence intervals (CIs) were estimated using multivariable Cox regression models. Analyses were also conducted for cancers associated with tobacco and alcohol after stratification for tobacco smoking and alcohol drinking.

RESULTS: Of the initial 142 605 men and 335 873 women included in the study, 9604 men and 21 000 women were identified with cancer after a median follow-up of 8.7 years. The crude cancer incidence rates were 7.9 per 1000 person-years in men and 7.1 per 1000 person-years in women. Associations between reduced cancer risk and increased intake of total fruits and vegetables combined and total vegetables for the entire cohort were similar (200 g/d increased intake of fruits and vegetables combined, HR = 0.97, 95% CI = 0.96 to 0.99; 100 g/d increased intake of total vegetables, HR = 0.98, 95% CI = 0.97 to 0.99); intake of fruits showed a weaker inverse association (100 g/d increased intake of total fruits, HR = 0.99, 95% CI = 0.98 to 1.00). The reduced risk of cancer associated with high vegetable intake was restricted to women (HR = 0.98, 95% CI = 0.97 to 0.99). Stratification by alcohol intake suggested a stronger reduction in risk in heavy drinkers and was confined to cancers caused by smoking and alcohol. **CONCLUSIONS:** A very small inverse association between intake of total fruits and vegetables and cancer risk was observed in this study. Given the small magnitude of the observed associations, caution should be applied in their interpretation.